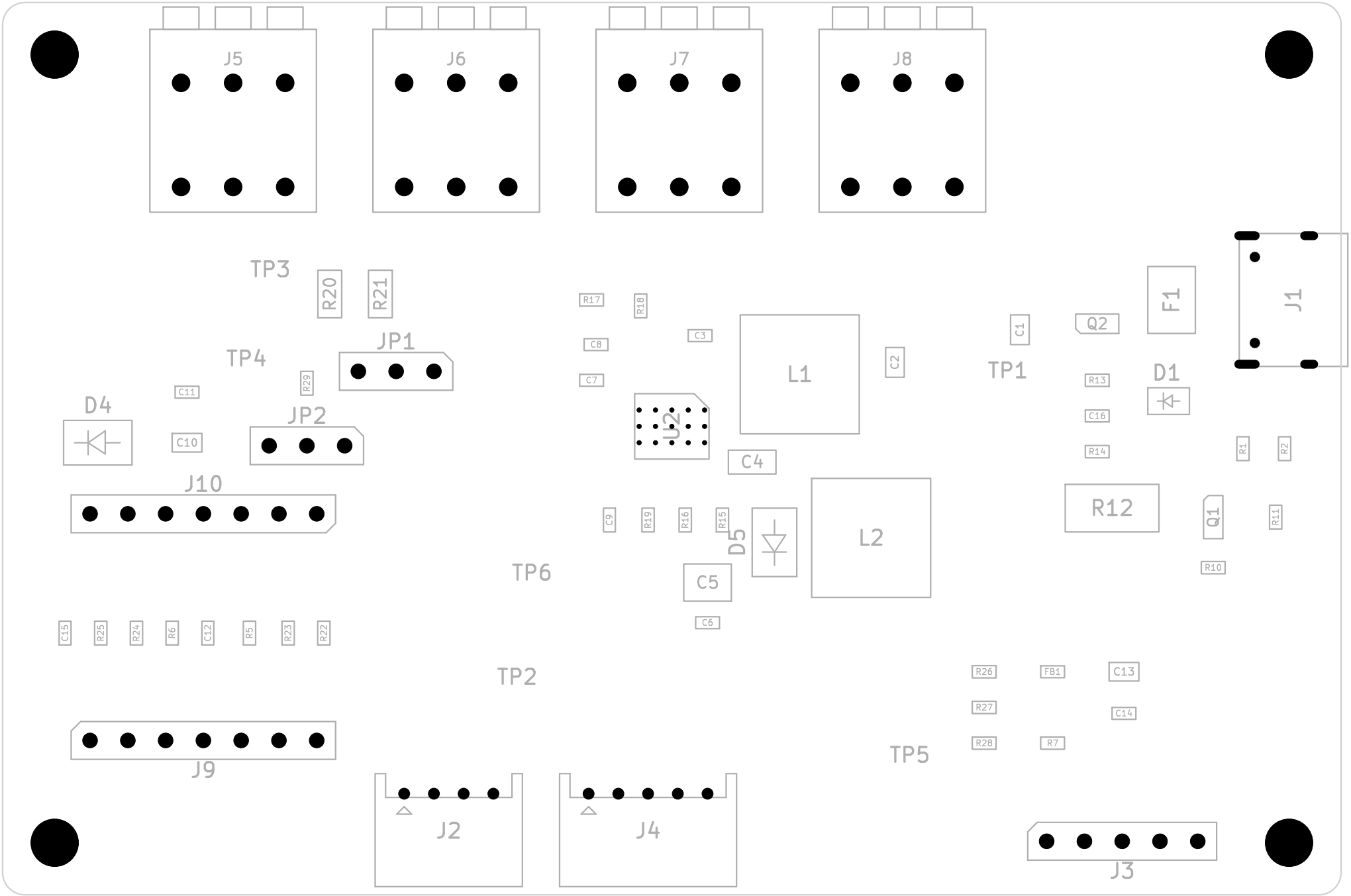
# Fused glass lighting controller Rev G2: assembly drawing, reference designators

F.Fab layer, 2026-10-29, 90 x 60 mm (D19). Parts under the XIAO and radar are hidden once those are plugged in. Top side, seen from above; about 3.0 : 1 on A4.



# Fused glass lighting controller Rev G2: parts by reference

From bom.csv, 2026-10-29. "Assembly" says who fits the part: JLCPCB, or you (plug-in modules, the JP1 and JP2 shunts).

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| **C1** | 10u 25V | CL21A106KAYNNNE | **J2** | POT 100k lin + SW | S4B-PH-K-S(LF)(SN) | **R10** | 100R 1% | 0603WAF1000T5E |
| **C2** | 10u 25V | CL21A106KAYNNNE | **J3** | LD2410S socket | HC-PM254-5.0H-1x5PZ | **R11** | 100k 1% | 0603WAF1003T5E |
| **C3** | 100n 50V | CC0603KRX7R9BB104 | **J4** | LD2410S cable | S5B-PH-K-S(LF)(SN) | **R12** | 33R 1% 1W | 25121WF330JT4E |
| **C4** | 10u 50V | CL31B106KBHNNNE | **J5** | PORT 1 | JL212R-35003B01 | **R13** | 100k 1% | 0603WAF1003T5E |
| **C5** | 10u 100V | GRM32EC72A106KE05L | **J6** | PORT 2 | JL212R-35003B01 | **R14** | 10k 1% | 0603WAF1002T5E |
| **C6** | 100n 100V | CL10B104KC8NNNC | **J7** | PORT 3 | JL212R-35003B01 | **R15** | 110k 1% | 0603WAF1103T5E |
| **C7** | 1u 50V | CL10A105KB8NNNC | **J8** | PORT 4 | JL212R-35003B01 | **R16** | 5.1k 1% | 0603WAF5101T5E |
| **C8** | 100n 50V | CC0603KRX7R9BB104 | **J9** | XIAO socket 1-7 | HC-PM254-5.0H-1x7PZ | **R17** | 1k 1% | 0603WAF1001T5E |
| **C9** | 100n 50V | CC0603KRX7R9BB104 | **J10** | XIAO socket 8-14 | HC-PM254-5.0H-1x7PZ | **R18** | 2.2k 1% | 0603WAF2201T5E |
| **C10** | 10u 25V | CL21A106KAYNNNE | **JP1** | ISEL | 2.54-1*3P | **R19** | 82k 1% | 0603WAF8202T5E |
| **C11** | 100n 50V | CC0603KRX7R9BB104 | **JP1-shunt** | 2.54 mm shunt for JP1 / JP2 | 2.54 [loose] | **R20** | 0.82R 1% | 1206W4F820LT5E |
| **C12** | 100n 50V | CC0603KRX7R9BB104 | **JP2-shunt** | 2.54 mm shunt for JP1 / JP2 | 2.54 [loose] | **R21** | 0.82R 1% | 1206W4F820LT5E |
| **C13** | 10u 25V | CL21A106KAYNNNE | **JP2** | RADAR_MODE | 2.54-1*3P | **R22** | 1k 1% | 0603WAF1001T5E |
| **C14** | 100n 50V | CC0603KRX7R9BB104 | **L1** | 10uH | SWPA8040S100MT | **R23** | 100k 1% | 0603WAF1003T5E |
| **C15** | 100n 50V | CC0603KRX7R9BB104 | **L2** | 10uH | SWPA8040S100MT | **R24** | 120k 1% | 0603WAF1203T5E |
| **C16** | 100n 50V | CC0603KRX7R9BB104 | **Q1** | AO3400A | AO3400A | **R25** | 10k 1% | 0603WAF1002T5E |
| **D1** | SMF6.0A | SMF6.0A [DNP] | **Q2** | AO3401A | AO3401A | **R26** | 1k 1% | 0603WAF1001T5E |
| **D4** | SS14 | SS14 | **R1** | 5.1k 1% | 0603WAF5101T5E | **R27** | 1k 1% | 0603WAF1001T5E |
| **D5** | SS36 | SS36 | **R2** | 5.1k 1% | 0603WAF5101T5E | **R28** | 1k 1% | 0603WAF1001T5E |
| **F1** | 2A hold | SMD1812B200TF/16 | **R5** | 47k 1% | 0603WAF4702T5E | **R29** | 10k | 0603WAF1002T5E |
| **FB1** | 600R@100MHz | GZ1608D601TF | **R6** | 1k 1% | 0603WAF1001T5E | **U1** | XIAO ESP32C3 | 113991054 [hand] |
| **J1** | USB-C (from bank) | TYPE-C-31-M-12 | **R7** | 100k 1% | 0603WAF1003T5E | **U2** | TPS61500PWPR | TPS61500PWPR |